engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element D Initial Template AT - 15 points

Because you will iterate (repeat) your design multiple times, you will need to fill out multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. Iteration must shine through in this Element.

After you have created an initial set of ideas, document your decision-making on which design to include with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1	*Composite based asphalt*
Date:	*October 21*
Description:	*Asphalt mixed with additives like polymers(plastic) for added flexibility and glass fibers to prevent cracking. While more costly than plain asphalt, this is cheaper in the long term as it is more structurally durable.*
Drawing that includes distinguishing information such as scale, dimensions, and materials:	*1.* *Asphalt is first milled down and coated with a tack coat that binds the new material*

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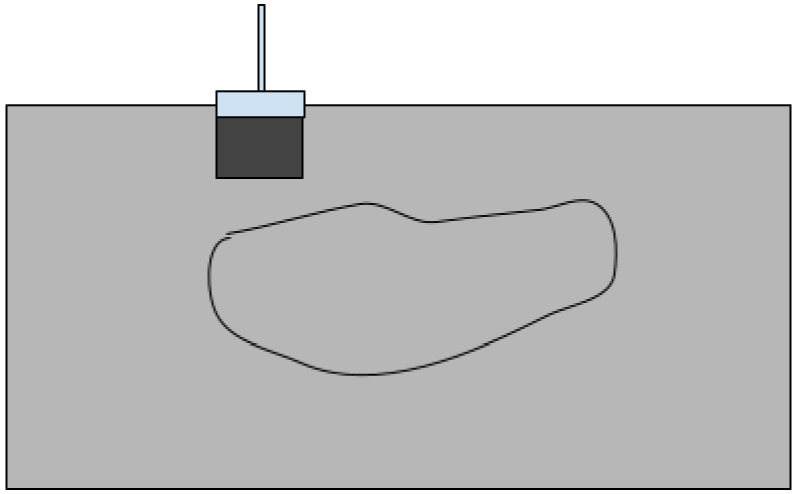
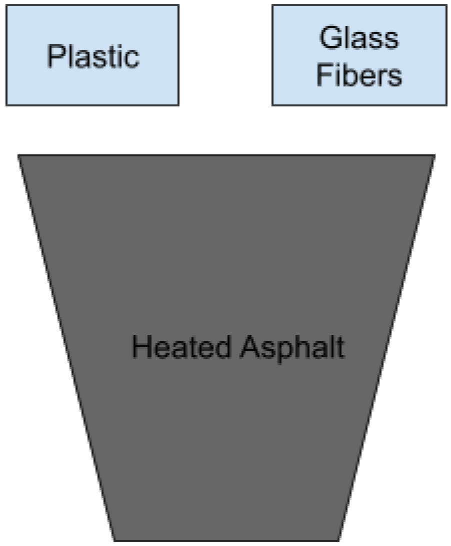
	 *2.* *Asphalt is first heated and mixed with glass fibers and polymers*  *3.* *The heated asphalt is poured into the hole*

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	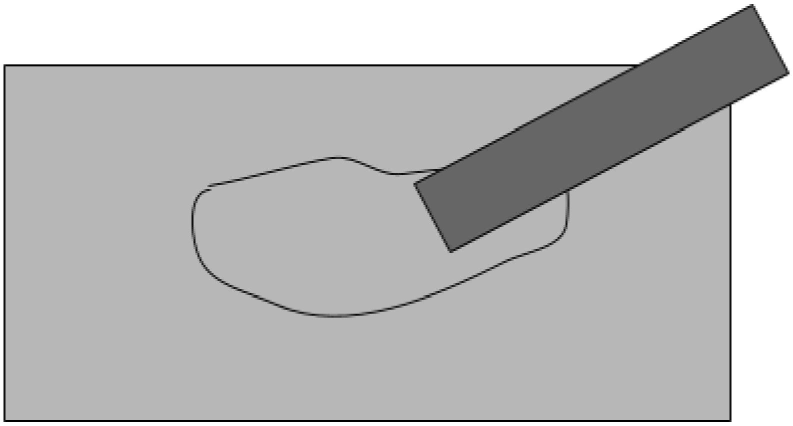

Design Idea #2	*Permeable Road Sample*
Date:	*October 18*
Description:	*There will be a one inch layer of sand and dirt layered below a half inch layer of permeable sediment to absorb the water and let it seep through, which collects at a 1 ½ inch tall detachable layer (using velcro) where the water is stored and can be emptied (replica of the sewage system). All of this will be stored in a 6x5x10 plastic container. This will prevent issues like flooding and cracks from freezing since the water will not sit between the cracks of a non-permable surface.*

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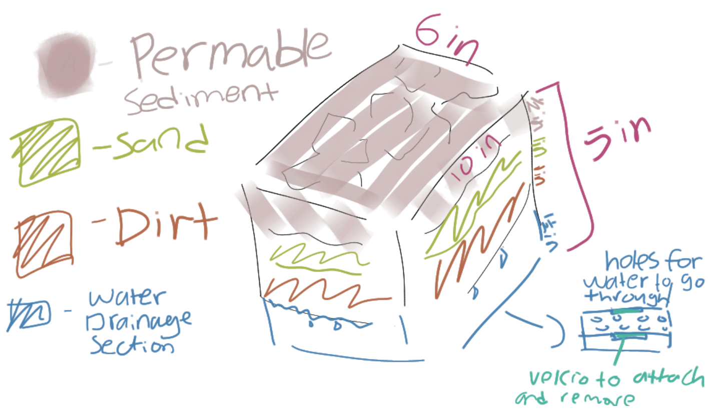
<tr><td>Drawing that includes distinguishing information such as scale, dimensions, and materials:</td><td>
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Design Idea #3	Porous sidewalk structure asphalt
Date:	*10/21/2025*
Description:	There will be three layers within the road with the the first and third being identical, the first will have a gravel block structure of lego blocks will have gaps in between, this will allow for water to flow through the asphalt with enough space to prevent flooding. This structure will also be much more stable for heavy vehicles to go through fast without the road breaking, the second layer is going to be a grid made of tar, this will allow flexible enough for asphalt to be stable but stiff enough to prevent the surface from splitting and potholes from forming. There will most likely be drainage systems underneath the road and just incase they flood during storms the asphalt above will not break the lego structure will compress and it will create and even hill like appearance on the road

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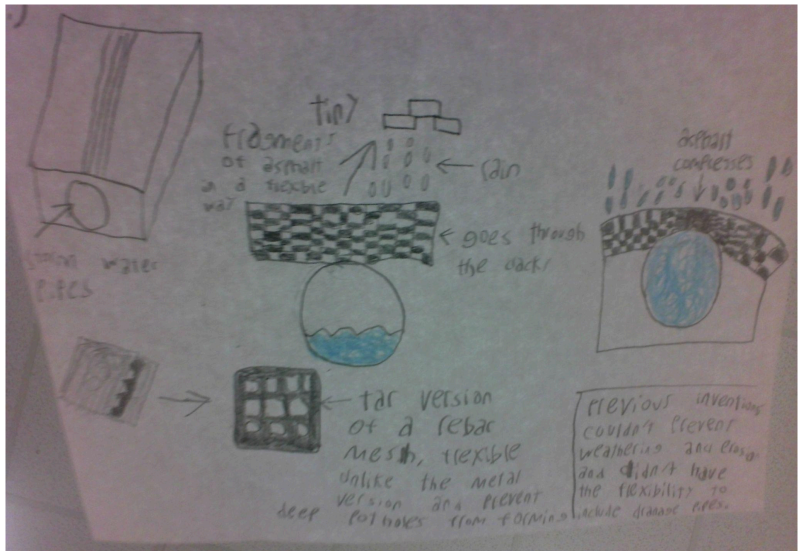
<table>
<tr><td></td><td>that will cause the rain to go sideways going into the sides, the lego like blocks of gravel on the side of the road will expand larger than their original state that will allow for more water to flow therefore replacing the closen gaps on the center of the road.</td></tr>
<tr><td>Drawing that includes distinguishing information such as scale, dimensions, and materials:</td><td>
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Design Idea #4	*4330 Bars W. Tarp*
Date:	*10/21*
Description:	*Bars made of 4330 Steel Alloy fitted to potholes & cracks, with tarp covering attached to seal off damage. Pipes made with internal lattice structure. The bars can be cut to fit the pothole, then arrayed to make a lattice structure throughout all 3 dimensions. They will be welded together, then a polyethylene tarp laid on top, cemented to joints & the pipes.*

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	If the pipes cannot be fit well enough, cement can be dumped into the pothole until there are exactly vertical walls & a horizontal base. Inside can be hollow (no need to flood entire lattice structure with cement.)
Drawing that includes distinguishing information such as scale, dimensions, and materials:	Cross Lattice Internal Structure  *2" to 4" diameter. Length can be cut to adjust to potholes or cracks. Tarp (polyethylene) also can be cut to fit cracks or potholes. Tarp is cemented to 4330 steel alloy skeleton.*

Design Idea Comparison and Selection *(D1, D2)*

(List your requirements from Element C below)	Weight (multiplied by each design score)	Design 1 *Composite based asphalt*	Design 2 *Permeable Road Sample*	Design 3 *Porous sidewalk structure asphalt*	Design 4 *Bars With Tarp*
Requirement A: Durability	1.5	1.5	1.5/1.5	1.5/1.5	1.5/1.5
Requirement B: Efficiency	1.5	1.5	1.5/1.5	1.5/1.5	1.5/1.5

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Requirement C: Quality	1.5	0	1.5/1.5	1.5/1.5	1.5/1.5
Requirement D: Aesthetics	1	1	1/1	1/1	1/1
Requirement E: Grip Texture	1	1	.5/1	.5/1	0/1
Total score	6.5	5/6.5	6/6.5	6/6.5	5.5/6.5

Our team chooses design #2 (Permeable Road Sample) to solve the community problem because it effectively allows water to seep through layers of permeable materials, reducing surface runoff, flooding, and cracking caused by trapped water on non-permeable surfaces. The detachable layer also provides a simple way to collect and remove excess water, replicating a sewage system and promoting sustainable water management.

Specific requirements it fills:

Design #2, the Permeable Road Sample, stands out because it meets several important requirements. It's durable since its layers of sand, dirt, and permeable sediment let water pass through instead of sitting on the surface, which helps prevent cracks and damage over time. It's also efficient, as the permeable layer quickly absorbs rainwater and channels it into a detachable storage section that can be easily emptied—similar to how a sewage system works. On top of that, the design shows strong quality because it's well thought out and uses materials that balance strength and function. Overall, this design is practical, long-lasting, and reliable, making it a smart way to reduce flooding and keep roads in better shape.

Design Blueprint *(D4)*

Describe all of your design details so that someone unfamiliar with your project can build your prototype.

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Include orthographic sketches (front, top, and side views), dimensions (include units), **CAD** drawings (if taught in your class.

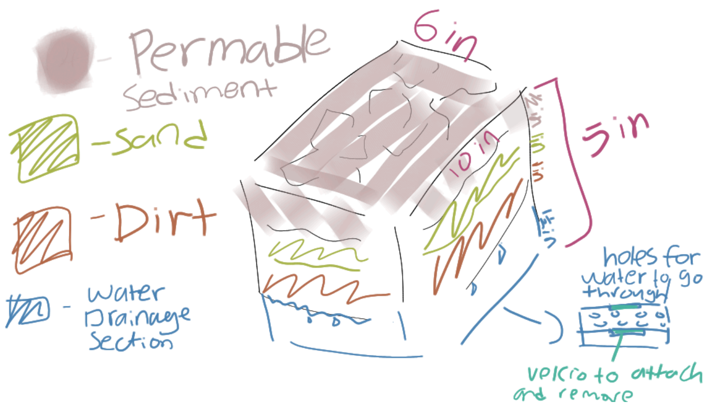
What manufacturing methods, material choices, supply lists, and costs do you anticipate for the prototype?

The materials would include:

1. Sand
2. Dirt
3. Permeable Sediment
4. Coarse gravel with air voids for drainage

Design Blueprint #1

Date:10/27
Sketch or CAD Drawing (orthographic):



Manufacturing Methods (cut, saw, paste, screw, tie, glue):
Step 1: get or make two containers around the size (6x4.5x10)(1.5x10x6) (there can be a difference of a inch from the original measurements)
Step 2: fill the (6x4.5x10) container with first a inch of dirt, then a inch of sand, then half a inch of permeable sediment.
Step 3: drill the (1.5x10x6) container with holes (diamerter of 1 cm, holes are one inch apart).

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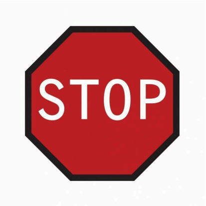
Step 4: apply velcro on the the top of the (1.5x10x6) container and the bottom of the (6x4.5x10) so they
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
Sand	9 dollars for cubic yard	60 cubic inches	$15
Dirt	$14 per cubic yard	60 cubic inches	$17
Permeable sediment	$2 per cubic yard	30 cubic inches	$1.50
Total Cost			$33.5

Plan of Action *(D3)*

- A Gantt chart is encouraged to outline the primary steps expected on a visual timeline. Here is one you can use.
    - Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document.
    - Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.



Wait until you create and test your first prototype to fill out the next section.

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration:

Description of Iteration:

Images if applicable:

Name: ______________________________ **Date:** _________________ **Class:** _______________

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions ______________.	was *comprehensive, iterative, and consistently* defensible (viable design	was *thorough, iterative, and* generally defensible (viable design likely)	was *adequate, generally* iterative and defensible *(viable design is possible)*	was *partial or overly general and only somewhat* iterative and/or defensible (raising issues with the design	*was incomplete or minimally iterative and/or defensible (raising issues with the design*	*was not evident that there was an attempt*

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	highly likely)			viability)	*viability)*	
2.Design Solution Choice. The Design Solution chosen ____________.	was *well-justified* and demonstrated attention to *all* design requirements	*was justified* and demonstrated attention to *most if not all* design requirements	*was explained with reference to at least some* design requirements	*was not sufficiently explained with reference to* design requirements	*lacked support related to* design requirements	*was not attempted or evidence not provided*
3. Plan of Action. The plan of action _____________.	*has considerable merit and would easily* support repetition and testing for effectiveness by others	*would* support repetition and testing for effectiveness by others	*might not clearly or fully support repetition and testing for effectiveness by others*	*has insufficient detail to allow for testing for replication of results*	*is lacking to the point that the design solution cannot be tested*	*cannot be executed or does not exist*